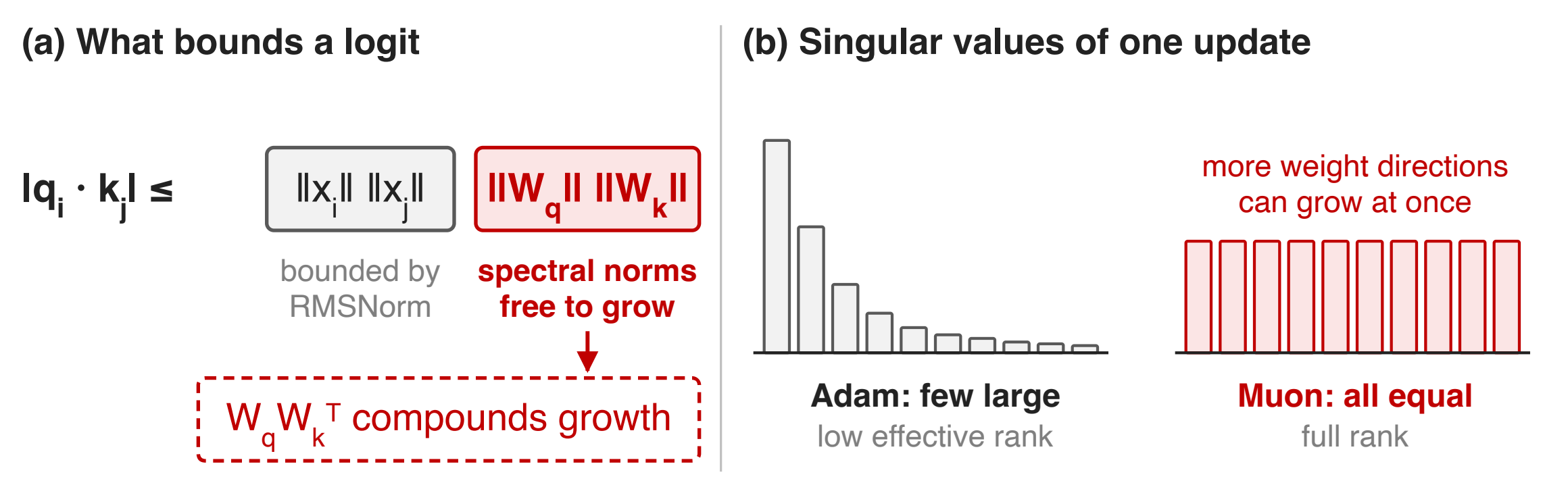

(a) What bounds a logit
$|q_i \cdot k_j| \leq$
$\|x_i\| \, \|x_j\|$
$\|W_q\| \, \|W_k\|$
bounded by RMSNorm
spectral norms free to grow
$W_q W_k^\top$ compounds growth
(b) Singular values of one update
more weight directions can grow at once
Adam: few large
low effective rank
Muon: all equal
full rank